Standard Operating Procedure

Battery-Operated Power Tools – Safety Procedure

Document No.	SOP-LV-007
Revision	1.0
Effective Date	July 13, 2026
Next Review Date	July 13, 2027
Applicable Jurisdiction(s)	Ontario (OHSA, O. Reg. 213/91) — Manitoba WSH Act/Regulation where applicable
Applies To	All field technicians, apprentices, and subcontractors operating battery-powered hand and power tools on Company work sites
Document Owner	Paul — Owner/Operator

1. Purpose

This SOP establishes safe work practices for the selection, inspection, use, charging, storage, and disposal of battery-operated (cordless) power tools used in low-voltage and structured cabling installation work, including drills, impact drivers, hammer drills, reciprocating saws, rotary tools, and related cordless equipment. It is intended to reduce the risk of electrical shock, laceration, eye injury, fire, and lithium-ion battery thermal events.

2. Scope

This procedure applies to all Company technicians, apprentices, and subcontracted personnel who operate, charge, transport, or store battery-operated power tools at commercial, institutional, or residential job sites, including active school sites during and outside instructional hours.

3. Definitions

- Thermal Runaway: An uncontrolled, self-heating chain reaction within a lithium-ion battery cell that can lead to fire or explosion.
- State of Charge (SOC): The current charge level of a battery relative to its full capacity.
- Damaged Battery: Any battery pack showing swelling, cracking, leakage, unusual odour, excessive heat, or impact damage.
- Competent Worker: A worker who is qualified by knowledge, training, and experience to perform the work and is familiar with the applicable regulations (per OHSA).

4. Responsibilities

Site Supervisor / Lead Technician

- Confirms all technicians using battery tools have received tool-specific and this SOP's training.
- Ensures damaged or recalled batteries/tools are removed from service immediately.
- Maintains a charging/storage area on-site that meets the requirements of Section 8.

Field Technician

- Performs pre-use inspection of tool and battery before each use.
- Wears all PPE specified in Section 6 for the task being performed.
- Reports any damaged battery, overheating, smoke, or unusual tool behaviour to the Site Supervisor immediately and stops work.

5. Approved Tools & Batteries

Only Company-issued or Company-approved battery-operated tools and OEM (or manufacturer-approved equivalent) battery packs and chargers may be used on Company work sites. Third-party or generic “compatible” battery packs are not permitted due to inconsistent cell protection circuitry and elevated fire risk.

6. Personal Protective Equipment (PPE)

- Drilling / driving: Safety glasses (CSA Z94.3), work gloves, hearing protection where sustained use >85 dB

- Cutting / reciprocating saw use: Safety glasses or face shield, cut-resistant gloves, hearing protection
- Overhead / drop ceiling work: Safety glasses, hard hat where overhead hazards exist, gloves
- Battery charging/handling: Safety glasses; gloves if a battery is suspected damaged or leaking

7. Pre-Use Inspection

Before each use, the technician must inspect and confirm:

- Tool housing, chuck, and trigger/safety switch are undamaged and function correctly.
- Battery pack contacts are clean, undamaged, and free of corrosion.
- Battery pack shows no swelling, cracking, leakage, discolouration, or unusual odour.
- Bits, blades, or attachments are correctly seated, sharp, and rated for the material being worked.
- Guards (where equipped) are in place and functional.

> ⚠️ **WARNING:**
> If any defect is found, remove the tool or battery from service and tag it out. Do not use a damaged battery under any circumstances.

8. Safe Operation

- Maintain full attention on the point of operation; keep hands and cords/hoses clear of bit or blade travel.
- Use two hands / auxiliary handle where provided, particularly for hammer drills and larger drivers.
- Be aware of kickback potential, especially in hammer-drill and rotary hammer modes; maintain firm footing and body position.
- Confirm the work area behind and around the drilling/cutting point is clear per SOP-LV-002 utility strike prevention procedures where applicable.
- Do not carry a tool by the battery pack, cord, or with a finger on the trigger.
- Remove the battery before changing bits, blades, or clearing jams.
- Do not operate tools with wet or oily hands, or in areas of standing water.
- Do not exceed the tool's rated capacity or use it for a task outside its intended purpose.

9. Battery Charging & Storage

Charging

- Charge batteries only in a designated area, on a non-combustible surface, away from flammable materials and combustible dust (e.g., away from ceiling tile debris, insulation, sawdust).
- Do not leave charging batteries unattended overnight or charge inside a closed vehicle in hot weather.
- Never charge a battery that is wet, frozen, damaged, or showing swelling.
- Charge within the manufacturer's specified temperature range; allow batteries to reach room temperature before charging if they have been in extreme heat or cold.

Storage & Transport

- Store batteries in a cool, dry location away from direct sunlight and heat sources.
- Transport batteries in a manufacturer case or with terminal caps/covers to prevent short-circuiting against metal tools or debris.
- Do not store or transport loose batteries in a pocket or tool bag alongside metal fasteners, screws, or bits.
- Segregate damaged or suspect batteries from serviceable stock and clearly tag as “OUT OF SERVICE.”

10. Damaged, Swollen, or Overheating Batteries

> ⚠️ **WARNING:**
> A swollen, leaking, smoking, or excessively hot battery is a fire hazard. Do not attempt to charge, use, or puncture it.

- Stop use immediately and, if safe to do so, move the battery outdoors or to a non-combustible, isolated location away from flammable materials.
- Do not attempt to extinguish a lithium-ion battery fire with water in its early stage if the cause is uncertain; use a Class D or ABC-rated extinguisher if trained and safe to do so, and evacuate if fire develops.

- Call 911 for any active fire; evacuate the area and account for all personnel.
- Notify the Site Supervisor immediately and complete an incident report consistent with Company incident reporting procedures.
- Dispose of damaged batteries only through a certified battery recycling/hazardous materials facility — never in general waste.

11. School & Occupied-Building Considerations

When working in active school or institutional environments (e.g., Elsie McGill Public School and similar sites):

- Keep charging stations and battery storage out of reach of students and away from high-traffic corridors.
- Do not leave tools, batteries, or chargers unattended in occupied areas; secure in a locked room, gang box, or vehicle between uses.
- Coordinate work timing and noise-generating tool use with the school board site contact where instructional activities are in progress.

12. Training Requirements

- All technicians must complete tool-specific manufacturer orientation before independent use of a new battery tool type.
- All technicians must review and acknowledge this SOP prior to first use of Company battery tools and at each annual review.
- Apprentices operate battery tools only under the direct supervision of a competent worker until deemed competent by the Site Supervisor.

13. Reference Standards

- Occupational Health and Safety Act (OHSA), Ontario
- O. Reg. 213/91 — Construction Projects
- CSA Z462 — Workplace Electrical Safety
- CSA Z94.3 — Eye and Face Protectors
- WSIB reporting requirements, Ontario
- Manitoba Workplace Safety and Health Act and Regulation, where work is performed in Manitoba
- Manufacturer's tool and battery operating instructions (govern where more restrictive than this SOP)

14. Related Documents

- SOP-LV-001 — Drop Ceiling Cat6 / Low-Voltage Installation
- SOP-LV-002 — Concrete Floor Drilling & Utility Strike Prevention
- SOP-LV-003 — Ontario Jobsite Health & Safety Checklist

Technician Acknowledgement & Sign-Off

By signing below, the technician confirms they have read, understood, and agree to follow this Standard Operating Procedure (SOP-LV-007) when using battery-operated power tools on Company work sites.

#	Printed Name	Signature	Date

#	Printed Name	Signature	Date

Reviewed By (Supervisor): ______________________ Signature: ______________________ Date: ______________________